

Neural Representation of Color in Color Vision Deficiency: An fMRI Study

Your Name

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Abstract

Placeholder abstract. Replace this with your actual abstract text. This study investigated neural color representations in individuals with color vision deficiency (CVD) using functional MRI and computational modeling approaches.

Keywords: color vision deficiency, fMRI, shared response model, forward encoding model, representational similarity analysis

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Methods**Participants**

Twelve volunteers were recruited under Institutional Review Board approval (#2510/002-023) and provided written informed consent. Color vision was screened using the 14-plate Ishihara pseudoisochromatic test (Ishihara, 1917). Participants identifying ≥ 13 plates correctly were classified as healthy controls (HC; $n = 7$, 3 female, age 22.7 ± 2.5 years). Two male participants scoring below this threshold were classified as CVD: sub-08 (deuteranomalous, 8/14 plates misread) and sub-09 (protanomalous, 11/14 plates misread). Sub-10 (2/14 plates misread; near-normal deutan) was retained as a within-study control. Three additional participants were excluded for failing to complete the full scanning protocol. All CVD results are interpreted as single-case demonstrations using the Crawford and Howell (1998) modified t -test; no population-level inference is made.

Stimuli and task

Eight isoluminant chromatic stimuli were drawn from CIE $L^*a^*b^*$ space at equal angular spacing (0° – 315° , 45° steps; $L^* = 75$, chroma = 40) (Commission Internationale de L'Eclairage, 1986), spanning red, orange, yellow, green, cyan, blue, purple, and magenta (Figure 1A). A gray filler ($a^* = b^* = 0$) served as a null condition.

Stimuli were presented in a Rapid Serial Visual Presentation (RSVP) design (Brouwer & Heeger, 2009) to sustain attention and fixation. A circular patch (1.5 s per stimulus) was shown in pseudo-random order optimised with Neurodesign (Durnez, Moerkerke & Nichols, 2018). Participants detected a consecutive-letter target embedded in a letter stream above the patch (button press on double 'K'). Each run contained eight repetitions of all eight hues (+ gray); inter-stimulus intervals were drawn from $\{3, 4.5, 6\}$ s. Each session comprised six runs (≈ 7 min each). The experiment was coded in PsychoPy 2022.2.5 (Peirce et al., 2019).

CVD participants completed a second session to evaluate the stimulus-space filters: two runs each under (i) unmodified stimuli, (ii) the 2-component-derived filter, and (iii) a generic Windows-accessibility filter, presented in counterbalanced order.

MRI acquisition and preprocessing

Data were acquired on a Siemens 3T MAGNETOM Cima.X scanner with BioMatrix 3T coils. To constrain coverage to posterior occipital cortex we used 24 oblique slices perpendicular to the calcarine sulcus (Ryu & Lee, 2024) (TR 1.5 s, TE 30 ms, FA 75°, $2 \times 2 \times 2$ mm, 96×80 matrix).

Data were converted to BIDS format and defaced with ezBIDS (Levitas et al., 2024). Functional images were registered to the T1w image via mutual-information coregistration (FreeSurfer `mri_coreg`; Maes, Collignon, Vandermeulen, Marchal & Suetens, 1997; Wells, Viola, Atsumi, Nakajima & Kikinis, 1996). T1w-to-MNI normalisation used 12-DOF affine (FSL FLIRT; Jenkinson, Bannister, Brady & Smith, 2002) followed by nonlinear warping (FNIRT; Andersson, Jenkinson & Smith, 2007), yielding 2 mm isotropic MNI-space BOLD data.

ROI definition and response estimation

Bilateral V1, V2, V3, and hV4 were defined from the Wang probabilistic atlas (Wang, Mruczek, Arcaro & Kastner, 2015) at $> 50\%$ probability, intersected with each participant’s BOLD brain mask (voxel counts: V1 655 ± 214 ; V2 451 ± 145 ; V3 103 ± 29 ; hV4 63 ± 22 ; mean $\pm$ SD).

Single-trial response amplitudes were estimated with a two-stage GLM (Dale, 1999; Brouwer & Heeger, 2009, 2013). First, an ROI-level hemodynamic response function (HRF) was extracted from a finite-impulse-response (FIR) model (8-TR basis set). Voxels in the top 50% of variance explained by the mean ROI HRF were retained (V1 328 ± 107 ; V2 226 ± 73 ; V3 52 ± 15 ; hV4 32 ± 11 voxels). Second, the estimated HRF was convolved with per-color design matrices and fitted by ridge regression to yield one amplitude per (voxel, color, run), organized as a $6 \times 8 \times V$ array per subject.

Voxel amplitudes were aligned across runs using within-subject Procrustes rotation (Gower, 1975): runs 2–6 were mapped to run 1 via the orthogonal transformation minimising $\|X_{\text{ref}} - XQ\|_F$ (scaling prohibited). This step preserves color representational geometry while attenuating run-to-run measurement noise.

Shared Response Model

Cross-subject alignment and dimensionality reduction were performed with the Shared Response Model (SRM; Chen et al., 2015; Haxby et al., 2011; Guntupalli et al., 2016). SRM estimates per-participant orthonormal mappings $W_i \in \mathbb{R}^{V \times k}$ and a group-shared response matrix $S \in \mathbb{R}^{k \times 8}$:

$$\min_{W_i, S} \sum_{i=1}^N \|X_i - W_i S\|_F^2 \quad \text{s.t.} \quad W_i^\top W_i = I_k \quad (1)$$

SRM was trained exclusively on the seven HC participants to avoid circular inference in CVD comparisons. Each CVD participant was subsequently projected into the fixed HC-derived space by solving $W_{\text{CVD}} = UV^\top$ where $U\Sigma V^\top = \text{SVD}(X_{\text{CVD}} \cdot S^\dagger)$. Reduced dimensions were chosen by mean-rank aggregation of leave-one-subject-out cross-validated LOCO performance: V1 $k = 4$; V2 $k = 4$; V3 $k = 3$; hV4 $k = 3$.

Discrimination decoding (LORO)

Categorical discrimination was assessed with leave-one-run-out (LORO) cross-validation in SRM-aligned space. For each of the six runs, a linear discriminant analysis (LDA) classifier was trained on the remaining five runs (mean-subtracted SRM projections; shrinkage regularisation) and tested on the held-out run. Performance was summarised as the 8-class exact accuracy (acc_exact; chance = $1/8 = 0.125$), averaged across six folds.

Group-level HC–CVD comparison used a Mann–Whitney U test on cross-subject generalization scores: the LDA model trained on each HC participant’s data was applied to every other participant, yielding 28 HC-to-HC and 12 HC-to-CVD pair-wise scores. The omnibus test (all ROIs pooled) characterised whether CVD subjects fell

outside the HC generalization distribution. Within-ROI single-case tests used the Crawford and Howell (1998) modified t -test (one-tailed, lower).

Interpolation decoding (LOCO) and vulnerability profile

Continuous hue interpolation was assessed with leave-one-color-out (LOCO) cross-validation using the forward encoding model (ForwardEncoding). Each of the 8 hues was held out in turn; the encoder was trained on the remaining 7 hues using all 6 runs (pooled 42-sample design matrix), and the held-out hue was predicted by nearest-neighbour matching in channel-output space.

Adjacent accuracy. The primary LOCO metric was adjacent accuracy (adj_acc): the proportion of predictions that fell within ± 1 hue step (45°) of the target hue, averaged across the 6 runs of each test color (range 0–1; chance level under a uniform random predictor = $3/8 = 0.375$; exact-accuracy chance = $1/8 = 0.125$). For HC above-chance testing and for pairwise HC–CVD comparisons at hV4, p -values were obtained from 1,000 random color-label permutations and from the Crawford and Howell (1998) modified t -test (one-tailed, lower), respectively.

Vulnerability profile. The 8-dimensional vector of per-hue adj_acc values, $\mathbf{v} \in [0, 1]^8$, constitutes each participant’s *hue vulnerability profile*. This profile is the phenotype inverted by the distortion model in Section . For group-level display, we additionally report Spearman ρ between predicted and observed $\mathbf{v}$ as the model-fit statistic (see below).

Per-hue group comparisons (Figure 2C) used the Crawford and Howell (1998) modified t -test (one-tailed, uncorrected); reported effect sizes are Hedges’ d with the Crawford–Howell correction for small- n comparisons.

Representational geometry: Δ RDM and disparity

To characterise the geometric basis of LOCO impairment we computed two complementary measures in SRM-aligned space.

Pairwise disparity. For each participant and ROI, the representational dissimilarity matrix (RDM) was computed as the 8×8 matrix of pairwise correlation

distances ($1 - \text{Pearson } r$). Mean pairwise disparity was defined as the average of all 28 upper-triangle elements. HC disparity was estimated with a leave-one-subject-out (LOO) procedure to remove the within-subject diagonal bias. Individual CVD disparity was compared to the HC LOO distribution using the Crawford and Howell (1998) modified t -test (one-tailed, upper; significance at $p < 0.05$).

ΔRDM . The deviation of CVD representational geometry from the HC mean was quantified as $\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC, LOO}}$, where $\overline{\text{RDM}}_{\text{HC, LOO}}$ is the HC mean RDM excluding the left-out subject. ΔRDM was visualised as a heatmap over all 28 color pairs (Figure 3A). To test whether a cone-shift model predicts the observed ΔRDM structure, we computed the Spearman ρ between the upper-triangle of the predicted $\Delta\text{RDM}_{\text{model}}$ and the observed ΔRDM , and assessed significance against 1,000 random sign-flip permutations of the upper-triangle elements.

Note that pairwise-disparity tests (Crawford & Howell) and ΔRDM permutation tests are statistically independent: the former tests whether absolute pairwise distances are elevated; the latter tests whether a specific distortion model accounts for the *structure* of those deviations. A significant pairwise-disparity finding does not imply a significant ΔRDM result and vice versa.

Two-component distortion model and parameter estimation

CVD hue distortion was modelled as a two-component angular shift in Stockman opponent-color space:

$$\delta\theta = \beta_s \cos(h - 90^\circ) + \beta_c \cos(h - \theta_{\text{conf}}) \quad (2)$$

where h is the opponent-hue angle of the stimulus and θ_{conf} is the CVD-subtype confusion axis (16° protan; 150° deutan). β_s parameterises distortion along the S-cone axis, motivated by behavioral evidence that CVD errors concentrate at S-cone loci (Emery, Volbrecht, Peterzell & Webster, 2021); β_c parameterises rotation along the confusion axis, capturing partial cortical compensation (Tregillus et al., 2021).

Opponent-hue angles were derived from the Machado simulation framework at zero

spectral shift (Machado, Oliveira & Fernandes, 2009).

Parameters were estimated by exhaustive grid search over $\beta_s \in [0^\circ, 50^\circ]$ and $\beta_c \in [-50^\circ, 50^\circ]$ ($26 \times 51 = 1,326$ combinations at 2° resolution). The pair $(\hat{\beta}_s, \hat{\beta}_c)$ maximising the Spearman ρ between the simulated and observed hV4 vulnerability profiles $\mathbf{v}$ was retained. Significance was assessed against $8! = 40,320$ exact permutations of color labels on the observed $\mathbf{v}$ (Figure 4B).

For comparison, a 1-DOF Machado cone-shift model ($\Delta\lambda$ only) and a 2-DOF retinal+cortical R+C model (Tregillus et al., 2021) were fitted identically. Model preference was based on the combination of (i) permutation-significant fit and (ii) exact pre-image existence for all 8 hues (see below).

Stimulus-space filter: pre-image computation

The personalized correction filter was derived as the pre-image of the fitted 2-component transform $T(\hat{\beta}_s, \hat{\beta}_c)$. For each of the 8 displayed hues θ_k , we solved for the input angle $\tilde{\theta}_k$ such that $T(\tilde{\theta}_k) = \theta_k$, yielding a per-hue correction $\delta\theta_k = \tilde{\theta}_k - \theta_k$. A CVD observer viewing $\tilde{\theta}_k$ thus receives the same cortical hue representation as a healthy control observer viewing θ_k .

Pre-images were computed numerically: 1,440 candidate values were evaluated on the forward map, a sign-change bracket was identified, and Brent’s method was applied for refinement. Bijectivity was confirmed by verifying $|T(\tilde{\theta}_k) - \theta_k| < 0.001^\circ$ for all 8 hues and both CVD participants. Because T is a smooth, single-valued circular function, a non-unique solution indicates arc compression (multiple inputs mapping to the same output) and inversion failure; this occurred for the Machado model for sub-09 (3 of 8 hues; Figure 5B) but not for the 2-component model.

Behavioral concordance

Participants completed two behavioral tasks under display conditions identical to the scanner experiment.

Just-noticeable difference (JND).. Color discrimination thresholds were estimated for eight hue pairs using two-interleaved 1-up/1-down adaptive staircases,

each terminating after eight reversals (Levitt, 1971). Pairs spanned three categories: (i) three pairs hypothesised as universally difficult for both CVD types (yellow–purple, blue–purple, red–orange), (ii) two pairs hypothesised as deutan-specific (orange–yellow, yellow–green), (iii) two pairs hypothesised as protan-specific (cyan–magenta, green–blue), and (iv) one control pair (red–cyan). LOCO–JND concordance was assessed by asking whether each pair’s LOCO-vulnerability classification (above vs. below HC range) agreed with the JND-impairment classification (above vs. below HC range).

8AFC color identification. Participants selected the matching hue from eight options identical to the scanner stimulus set. Same-day performance confirmed Ishihara classifications (reported in Section).

Reproducibility

All analyses were conducted in Python 3.10 (numpy 1.24.3, scipy 1.11.3, scikit-learn 1.3.0, BrainIAK 0.11). Random seeds were fixed (`seed=42`) for all permutation tests and bootstrap resampling. Analysis code and preprocessing scripts are available at https://github.com/haba6030/colorBlind_analysis. Anonymised preprocessed data are available on reasonable request to the corresponding author.

Results

Participants and behavioral phenotype

Two CVD participants spanning the deutan–protan dichotomy were recruited alongside seven healthy controls (HC; sub-01–07). Sub-08 (male) was classified as mild–moderate deuteranomalous (8/14 Ishihara plates misread; deutan axis confirmed by protan–deutan discrimination item); Sub-09 (male) as moderate–severe protanomalous (11/14 plates misread; protan axis). Same-day 8-alternative forced-choice (8AFC) identification of the eight scanner hue stimuli confirmed the Ishihara classification (HC: 0.94 ± 0.04 ; Sub-08: 0.71; Sub-09: 0.62; chance = 0.125). Sub-08’s errors clustered on the red–green axis (c2/c3 vs. c7; mean confusion 28%);

Sub-09’s errors extended across the L–M opponent range with additional misidentification of magenta (c8). Because only two symptomatic CVD cases were available, all inferences are framed as single-case analyses following Crawford and Howell (1998) and Schütt, Kipnis, Diedrichsen and Kriegeskorte (2023); no population-level CVD claim is made. The experimental paradigm and full analysis pipeline are summarised in Figure 1.

Color discrimination is preserved across all ROIs in CVD

A corrective stimulus-space filter is meaningful only if the visual cortex of a CVD individual retains categorical discrimination among the target hues. We verified this precondition using leave-one-run-out (LORO) cross-validation of a linear discriminant classifier applied to SRM-aligned responses.

Both CVD participants exceeded the 0.125 exact-accuracy chance level at every ROI (Figure 2A). A cross-subject LDA generalization test (HC-to-HC vs. HC-to-CVD pair-wise cross-decoding, all ROIs pooled) showed no HC–CVD difference ($p = 0.668$, Mann–Whitney U , 28 HC-to-HC vs. 12 HC-to-CVD pairs). The same null held at the within-ROI level at hV4 (Crawford & Howell t -test: both CVD subjects within the HC range; $p = 0.142$). In line with prior work on CVD above-threshold identification (Bosten, 2019; Boehm, MacLeod & Bosten, 2014), the 8 scanner stimuli are individually distinguishable at the cortical level in both CVD participants; the representational substrate on which a corrective filter could act is present.

hV4 color interpolation is selectively impaired in CVD

We next asked where continuous hue geometry fails — the functional target of the corrective transform. Leave-one-colour-out (LOCO) decoding trains the encoder on 7 of 8 hues and measures whether the held-out hue is predicted from the remaining representational structure, yielding an 8-dimensional vulnerability profile per participant.

Among the four retinotopic ROIs, only hV4 supported above-chance hue interpolation in HC (adjacent accuracy $\text{adj_acc} = 0.47 \pm 0.05$, mean $\pm$ SEM;

$p_{\text{perm}} = 0.044$, label permutation; V1–V3 did not exceed chance despite preserving LORO discrimination). This confirms hV4 as the primary interpolation gate, replicating Brouwer and Heeger (2009).

Both CVD participants fell at or below the adjacent-accuracy chance level at hV4 (Sub-08: $\text{adj_acc} = 0.25$, Crawford & Howell $t = -1.58$, $p = 0.082$, n.s.; Sub-09: $\text{adj_acc} = 0.13$, $t = -2.48$, $p = 0.024$; Figure 2B). The impairment was not uniform across hues: both participants showed near-zero adjacent accuracy specifically for S-cone intermediate colors (blue, purple, magenta at hV4; per-hue Crawford & Howell tests: blue, Sub-08 $d = 2.13$, $p = 0.047$; Sub-09 $d = 2.15$, $p = 0.046$; purple, Sub-08 $d = 2.40$, $p = 0.033$; Figure 2C). This per-hue impairment pattern is consistent with both participants’ behavioral error distributions from the 8AFC task and with the JND confusion pairs identified in the pre-scanner session (LOCO–JND concordance 100%, 6/6 pairs; SRM geometry–JND concordance 33%, 2/6 pairs), establishing LOCO as the functional phenotype that predicts behavioral color discrimination failure.

CVD color representation geometry is distorted in a subject-specific, ROI-specific pattern

To characterise the geometric basis of the LOCO impairment, we compared the pairwise representational structure of CVD and HC participants in a shared SRM-aligned space. For each CVD participant we computed the RDM difference ($\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC,LOO}}$) and the mean pairwise correlation distance (disparity).

Sub-08 showed significantly elevated disparity specifically at V2 (Crawford & Howell $p = 0.040$), with no significant elevation at V1, V3, or hV4. Sub-09 showed significantly elevated disparity specifically at V1 ($p = 0.007$). Sub-10 (near-normal deutan, included as a within-study control) showed no significant elevation at any ROI (all $p > 0.43$), consistent with the null behavioral phenotype. The ROI specificity diverges between the two CVD participants — V2-dominant for deutan, V1-dominant for protan — and is inconsistent with a shared group-level gain mechanism (Figure 3B).

The Δ RDM heatmaps (Figure 3A) reveal subject-specific distortion structures: Sub-08 shows elevated distances among S-cone intermediate pairs at V2; Sub-09 shows a broader reorganisation of pairwise distances at V1. A permutation test of the retinal+cortical (R+C) cone-shift model against the observed Δ RDM was significant for Sub-09 at V1 ($p = 0.026$) but not Sub-08 at V2 ($p = 0.179$), indicating that geometric distortion measured at the population-code level is complementary to, not redundant with, the LOCO-based functional characterisation.

A two-component model captures individual CVD hV4 LOCO vulnerability

To identify the distortion parameters needed to build a corrective filter, we fitted three nested models to each participant’s hV4 LOCO vulnerability profile by grid search: (i) the 1-DOF Machado cone shift ($\Delta\lambda$; Machado et al., 2009), (ii) the 2-DOF cortical 2-component model (β_s for S-cone axis, β_c for confusion-axis rotation), and (iii) the 2-DOF retinal+cortical R+C model ($\Delta\lambda, g$; Tregillus et al., 2021).

The 2-component model reached significance for both participants (Sub-08: $\beta_s = 38^\circ$, $\beta_c = -14^\circ$, Spearman $\rho = 0.88$, $p_{\text{perm}} = 0.004$; Sub-09: $\beta_s = 6^\circ$, $\beta_c = -22^\circ$, $\rho = 0.69$, $p = 0.035$; 40,320 permutations; Figure 4B). Machado reached significance for Sub-09 ($\Delta\lambda = 13.5$ nm, $\rho = 0.76$, $p = 0.018$) but not for Sub-08 ($\rho = 0.62$, $p = 0.058$, n.s.). The 2-component model is preferred for both subjects because it yields exact pre-image solutions for all 8 hues (see Section), whereas Machado produces arc compression that prevents exact inversion for Sub-09.

The parameter landscape (Figure 4C) reveals a clear ridge structure: β_s and β_c trade off along a diagonal for each subject, with distinct optimal locations. The β_s values (38° for Sub-08; 6° for Sub-09) are consistent in range with the 21.4° S-cone phase shift reported by Emery et al. (2021) in an independent anomalous-trichromat cohort assessed via behavioral hue scaling; however, direct numerical comparison is not warranted as the quantities are defined on different physical scales (degrees of DKL angular rotation vs. behavioral scaling units). The β_s values serve here as physiological grounding for the model structure, not as parameter-value convergence claims.

Personalized stimulus-space filters are exact and individually distinct

We computed the per-hue correction filter as the stimulus-space pre-image of the 2-component distortion. For each of the 8 displayed hues we solved for the input angle θ_{in} such that the 2-component forward map produces the target hue θ_{target} , yielding a correction vector $\delta\theta \in \mathbb{R}^8$ per participant.

Both subjects yielded exact pre-images for all 8 hues (maximum residual $< 0.001^\circ$; Figure 5A). The per-subject mean correction magnitudes differ substantially: Sub-08 requires a mean shift of 46.3° (maximum 104.2° at cyan); Sub-09 requires 20.1° (maximum 48.1° at orange), a $2.3\times$ difference in correction scale.

For comparison, the Machado pre-image for Sub-09 collapses three hues (green 135° , cyan 180° , blue 225°) onto a single pre-image angle ($\sim 127^\circ$), preventing exact inversion (4/8 exact; Figure 5B). The 2-component model, whose inverse is guaranteed to exist by construction (bijective angular dilation), resolves this arc compression and yields exact corrections at all 8 hue positions.

The two subjects' correction profiles differ not only in magnitude but also in sign at 4 of 8 hue positions (cyan–magenta arc, hues 5–8; Figure 5C; sub-08 vs. sub-09 filter cosine similarity = 0.55). A correction optimised for one participant's deutan geometry would move orange, cyan, blue, purple, and magenta in the *wrong* direction for the other participant's protan geometry. This divergence directly motivates the individual-derivation approach and disqualifies any group-average correction for this pair of subjects.

References

- Andersson, J. L., Jenkinson, M. & Smith, S. (2007). *Non-linear registration aka spatial normalisation* (Technical Report No. TR07JA2). FMRIB.
- Bannert, M. M. & Bartels, A. (2025). Color across human brains: Shared cortical representation of color in healthy observers. *Journal of Neuroscience*, *45*(42).
- Benjamini, Y. & Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, *57*(1), 289–300. doi: doi:10.1111/j.2517-6161.1995.tb02031.x
- Boehm, A. E., MacLeod, D. I. A. & Bosten, J. M. (2014). Compensation for red-green contrast loss in anomalous trichromats. *Journal of Vision*, *14*(13), 19. doi: doi:10.1167/14.13.19
- Bosten, J. M. (2019). The known unknowns of anomalous trichromacy. *Current Opinion in Behavioral Sciences*, *30*, 228–237. doi: doi:10.1016/j.cobeha.2019.10.015
- Brouwer, G. J. & Heeger, D. J. (2009). Decoding and reconstructing color from responses in human visual cortex. *Journal of Neuroscience*, *29*(44), 13992–14003. doi: doi:10.1523/JNEUROSCI.3577-09.2009
- Brouwer, G. J. & Heeger, D. J. (2013). Categorical clustering of the neural representation of color. *Journal of Neuroscience*, *33*(39), 15454–15465. doi: doi:10.1523/JNEUROSCI.2472-13.2013
- Chen, P. H., Chen, J., Yeshurun, Y., Hasson, U., Haxby, J. & Ramadge, P. J. (2015). A reduced-dimension fmri shared response model. In *Advances in neural information processing systems* (Vol. 28, pp. 460–468).
- Commission Internationale de L’Eclairage. (1986). *Colorimetry* (2nd ed.). CIE. (Publication No. 15.2)
- Crawford, J. R. & Howell, D. C. (1998). Comparing an individual’s test score against norms derived from small samples. *The Clinical Neuropsychologist*, *12*(4), 482–486. doi: doi:10.1076/clin.12.4.482.7241
- Cunningham, J. P. & Yu, B. M. (2014). Dimensionality reduction for large-scale neural

- recordings. *Nature Neuroscience*, 17(11), 1500–1509. doi: doi:10.1038/nn.3776
- Dale, A. M. (1999). Optimal experimental design for event-related fmri. *Human Brain Mapping*, 8(2-3), 109–114. doi: doi:10.1002/(SICI)1097-0193(1999)8:2/3<109::AID-HBM7>3.0.CO;2-W
- Deeb, S. S. (2005). The molecular basis of variation in human color vision. *Clinical Genetics*, 67(5), 369–377. doi: doi:10.1111/j.1399-0004.2004.00343.x
- Durnez, J., Moerkerke, B. & Nichols, T. E. (2018). neurodesign: optimal experimental designs for task fmri. *Journal of Open Source Software*, 3(32), 564. doi: doi:10.21105/joss.00564
- Emery, K. J., Volbrecht, V. J., Peterzell, D. H. & Webster, M. A. (2021). Color perception and compensation in color deficiencies assessed with hue scaling. *Vision Research*, 183, 1–15. doi: doi:10.1016/j.visres.2021.01.006
- Freeman, W. T. & Adelson, E. H. (1991). The design and use of steerable filters. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 13(9), 891–906. doi: doi:10.1109/34.93808
- Golub, G. H., Heath, M. & Wahba, G. (1979). Generalized cross-validation as a method for choosing a good ridge parameter. *Technometrics*, 21(2), 215–223. doi: doi:10.1080/00401706.1979.10489751
- Gower, J. C. (1975). Generalized procrustes analysis. *Psychometrika*, 40(1), 33–51. doi: doi:10.1007/BF02291478
- Guntupalli, J. S., Hanke, M., Halchenko, Y. O., Connolly, A. C., Ramadge, P. J. & Haxby, J. V. (2016). A model of representational spaces in human cortex. *Cerebral Cortex*, 26(6), 2919–2934. doi: doi:10.1093/cercor/bhw068
- Haxby, J. V., Guntupalli, J. S., Connolly, A. C., Halchenko, Y. O., Conroy, B. R., Gobbini, M. I., ... Ramadge, P. J. (2011). A common, high-dimensional model of the representational space in human ventral temporal cortex. *Neuron*, 72(2), 404–416. doi: doi:10.1016/j.neuron.2011.08.026
- Ishihara, S. (1917). *Tests for colour-blindness*. Tokyo: Kanehara Shuppan.
- Jenkinson, M., Bannister, P., Brady, M. & Smith, S. (2002). Improved optimization for

- the robust and accurate linear registration and motion correction of brain images. *NeuroImage*, 17(2), 825–841. doi: doi:10.1016/s1053-8119(02)91132-8
- Levitas, D., Hayashi, S., Vinci-Booher, S., Heinsfeld, A., Bhatia, D., Lee, N., ... Pestilli, F. (2024). ez bids: Guided standardization of neuroimaging data interoperable with major data archives and platforms. *Scientific Data*, 11, 179. doi: doi:10.1038/s41597-024-02959-0
- Levitt, H. (1971). Transformed up-down methods in psychoacoustics. *The Journal of the Acoustical Society of America*, 49(2), 467–477. doi: doi:10.1121/1.1912375
- Machado, G. M., Oliveira, M. M. & Fernandes, L. A. F. (2009). A physiologically-based model for simulation of color vision deficiency. *IEEE Transactions on Visualization and Computer Graphics*, 15(6), 1291–1298. doi: doi:10.1109/TVCG.2009.113
- Maes, F., Collignon, A., Vandermeulen, D., Marchal, G. & Suetens, P. (1997). Multimodality image registration by maximization of mutual information. *IEEE Transactions on Medical Imaging*, 16(2), 187–198. doi: doi:10.1109/42.563664
- Neitz, J. & Neitz, M. (2011). The genetics of normal and defective color vision. *Vision Research*, 51(7), 633–651. doi: doi:10.1016/j.visres.2010.12.002
- Nichols, T. E. & Holmes, A. P. (2002). Nonparametric permutation tests for functional neuroimaging: a primer with examples. *Human Brain Mapping*, 15(1), 1–25. doi: doi:10.1002/hbm.1058
- Peirce, J., Gray, J. R., Simpson, S., MacAskill, M., Höchenberger, R., Sogo, H., ... Lindeløv, J. K. (2019). Psychopy2: Experiments in behavior made easy. *Behavior Research Methods*, 51(1), 195–203. doi: doi:10.3758/s13428-018-01193-y
- Ryu, J. & Lee, S.-H. (2024). Slice planning perpendicular to calcarine sulcus for retinotopic mapping. *Journal of Neuroscience Methods*. (In preparation)
- Schütt, H. H., Kipnis, A. D., Diedrichsen, J. & Kriegeskorte, N. (2023). Statistical inference on representational geometries. *eLife*, 12, e82566. doi: doi:10.7554/eLife.82566
- Tregillus, K. E. M., Isherwood, Z. J., Vanston, J. E., Engel, S. A., MacLeod, D. I. A.,

- Kuriki, I. & Webster, M. A. (2021). Color compensation in anomalous trichromats assessed with fmri. *Current Biology*. doi: doi:10.1016/j.cub.2021.01.039
- Wang, L., Mruczek, R. E., Arcaro, M. J. & Kastner, S. (2015). Probabilistic maps of visual topography in human cortex. *Cerebral Cortex*, 25(10), 3911–3931. doi: doi:10.1093/cercor/bhu277
- Wells, W. M., Viola, P. A., Atsumi, H., Nakajima, S. & Kikinis, R. (1996). Multi-modal volume registration by maximization of mutual information. *Medical Image Analysis*, 1(1), 35–51. doi: doi:10.1016/S1361-8415(01)80004-9

Supplementary Methods

S1. Confound Regression and Temporal Filtering

Within-run head motion was estimated via rigid-body realignment. No temporal filtering or confound regression was applied; slow drift was modeled via linear per-run regressors in the general linear model.

S2. Quality Control

Registration quality was evaluated by computing the intersection between the Wang Atlas ROIs (V1, V2, V3, hV4) and the BOLD brain mask in MNI space. Across 10 subjects, mean ROI coverage was 84.3% (SD = 21.7%), and GLM valid ratio (percentage of ROI voxels with reliable stimulus-evoked responses) was 99.6%. We evaluated all subjects' data suitable for downstream analysis, though sub-07 showed reduced coverage (30.8%) due to individual anatomical variability.

S3. Voxel Counts Range

Voxel counts after ROI definition and quality filtering are reported in the main Methods section. Detailed per-subject counts are available in the online repository.

S4. Consistency Test: PCA and Procrustes (TBA)

Additional validation analyses comparing SRM with PCA-based dimensionality reduction and norm-constrained Procrustes alignment will be reported following completion of the full dataset.

S5. Mean Activation Analysis

We conducted mean voxel response amplitudes analysis to identify whether SRM-based differences rely on the activation. We checked mean amplitudes across all colors and for each color.

S6. Statistical Analysis

All tests used a significance threshold of $\alpha = 0.05$ with False Discovery Rate correction (Benjamini & Hochberg, 1995); $q = 0.05$ applied to RDM color-pair comparisons (28 pairs per participant). Individual CVD participants were compared to the HC distribution using Crawford and Howell (1998) modified t-tests:

$$t^* = \frac{x_{\text{CVD}} - \bar{x}_{\text{HC}}}{\text{SD}_{\text{HC}} \times \sqrt{(n+1)/n}}, \quad df = n - 1 = 6 \quad (3)$$

with one-tailed p-values testing the directional hypothesis that CVD disparity exceeds HC. Effect sizes were quantified via Hedges' g . Group-level permutation tests (10,000 iterations) were used for HC-CVD comparisons by permuting subject labels; color-level permutation tests (1,000 iterations) were used for LOCO/LORO validation by permuting color labels within subjects.

S7. Dimensionality Selection (K-Selection)

The reduced dimension k for SRM was determined via leave-one-subject-out (LOSO) cross-validation based on within-subject representational dissimilarity matrix (RDM) reliability (Cunningham & Yu, 2014). For each candidate k (range: 1–8) and fold, reliability was quantified as the Spearman correlation between RDMs computed from even runs (2,4,6) versus odd runs (1,3,5) of the held-out participant in SRM space. We selected k maximizing mean reliability across seven folds. The selected values and their associated cross-subject RDM Spearman correlations were:

- V1: $k = 4$ (cross-subject RDM correlation = 0.60)
- V2: $k = 4$ (cross-subject RDM correlation = 0.57)
- V3: $k = 3$ (cross-subject RDM correlation = 0.55)
- hV4: $k = 3$ (cross-subject RDM correlation = 0.32)

S8. Generalized Cross-Validation for Ridge Regression

The regularization parameter α for ridge regression was selected via Generalized Cross-Validation (GCV; Golub, Heath & Wahba, 1979). GCV provides a computationally efficient approximation to leave-one-out cross-validation for selecting α . The optimal α minimizes the GCV score:

$$\text{GCV}(\alpha) = \frac{1}{n} \|y - \hat{y}\|_2^2 / \left(1 - \frac{\text{tr}(H)}{n}\right)^2 \quad (4)$$

where X denotes the design matrix, y is the observed response vector, $\hat{y}$ is the ridge-predicted response, $H = X(X^T X + \alpha I)^{-1} X^T$ is the hat matrix, and n is the number of training samples.

S9. Leave-One-Out Consistent Disparity Estimation

To quantify HC-CVD disparity with unbiased references, we used a leave-one-out (LOO) consistent procedure: for each fold i (leaving out HC participant i), a reference pattern R_{-i} was computed as the mean of the remaining six HC participants' aligned data. Both the held-out HC_i and both CVD participants were compared against this same reference R_{-i} , yielding disparity scores $d(\text{HC}_i, R_{-i})$ and $d(\text{CVD}_j, R_{-i})$. This ensures HC and CVD disparities are measured against identical references within each fold. Individual CVD disparity significance was assessed via Crawford and Howell (1998) modified t-tests comparing each CVD participant's mean disparity (averaged across 7 folds) to the HC distribution ($n = 7$, $df = 6$).

S10. Cross-Validation Procedures

Leave-One-Run-Out (LORO)

LORO evaluated whether individual colors could be distinguished from voxel response amplitudes, testing the consistency of neural color encoding across runs. For each of six runs, the corresponding response matrices (B , C) were held out as the test set, W was trained on the remaining five runs. Both color decoding and voxel response prediction were evaluated on the held-out run.

Leave-One-Color-Out (LOCO)

LOCO tested whether geometric relationships of circular hue space were preserved in neural representation by evaluating interpolation to held-out colors. For each color, all six run response matrices of the target color were held out. W was trained on the remaining seven colors, and the model predicted either the held-out color from voxel responses or voxel responses from the held-out color’s channel outputs.

Leave-One-Subject-Out (LOSO)

LOSO evaluated zero-shot generalization to held-out participants. As LOSO requires cross-subject prediction, SRM-transformed data were used to align voxel spaces across participants. For each HC participant, W was trained on the remaining HC participants’ shared-space data and tested on the held-out participant. This assessed whether the HC group-derived common space supported generalization to novel individuals.

S11. Evaluation Metrics**Color Decoding Metrics**

Color decoding was evaluated using (1) mean absolute error (MAE) between predicted and true hue angles (chance = 90° , expected from a uniform random predictor on a circle), and (2) categorization accuracy with $\pm 22.5^\circ$ bins around each of the eight colors (chance = $12.5\% = 1/8$).

Voxel Prediction Metrics

Voxel response prediction was evaluated using Pearson correlation between predicted and observed response vectors, computed across all voxels and colors in the test set.

Statistical Significance

Statistical significance of group comparisons (HC vs CVD) and individual CVD analyses was assessed using nonparametric permutation tests (10,000 iterations; Nichols & Holmes, 2002).

S12. Shared Response Model Residuals

Each participant’s residuals in the SRM model reflect the extent to which their data are not captured by the common structure, including inherent geometric differences. The orthogonal constraint ($W_i^T W_i = I_k$) helps preserve geometric relationships between colors—distances and angles—in the common space.

Additionally, global between-subject RDM similarity was quantified using Spearman correlation across all 28 color pairs.

S13. Image Orientation Initialization

Image orientation was first initialized according to the scanner-defined obliquity information in the NIfTI header. The mutual-information cost function was then optimized to maximize the statistical dependence between T1w and BOLD intensity distribution within the overlapping region.

S14. Procrustes Alignment Benefits

This preprocessing reduces measurement noise across runs while preserving color representational geometry, thereby improving the quality of subsequent SRM fitting.

S15. HC Specificity Calibration

The 2-component grid-search pipeline was applied to the 7 HC participants treating each as a CVD candidate. Of 21 model–subject tests ($7 \text{ HC} \times 3 \text{ models}$), 15 reached nominal $p < 0.05$ (71%). This elevated false-positive rate reflects (i) rank-based loss with only 8 data points (Spearman ρ over 8 hues is sensitive to individual rank fluctuations), and (ii) regression to the mean on per-HC baseline ρ (hV4 $K = 3$, 63 voxels; individual HC baselines span $[-0.36, +0.69]$). Case-level specificity therefore

rests on the three converging anchors described in ?? , not on per-subject permutation significance alone.

S16. Comparison with Retinal-Family Distortion Models

Two retinal-family models were evaluated alongside the 2-component model by the same grid-search and permutation procedure (see ??).

Models. The 1-DOF Machado model Machado et al., 2009 shifts the L/M cone spectral sensitivity by $\Delta\lambda$ nm, yielding a hue-confusion pattern characteristic of the CVD subtype. The 2-DOF retinal+cortical R+C model Tregillus et al., 2021 augments Machado with a cortical gain g : $r'_g = r_{g,\text{base}} + (1 + g)(r_{g,\text{ret}} - r_{g,\text{base}})$, where $g = 0$ recovers the retinal-only model and $g = -1$ is exact cortical compensation.

Fits (Table 1). Applying a closed-testing rule (Machado first; R+C escalated only if Machado $p > 0.05$), the retinal family reaches significance for both subjects but selects a different model class per case. Sub-08 escalates to R+C ($\Delta\lambda = 2.0$ nm, $g = +2.25$, $\rho = 0.857$, $p = 0.005$) because 1-DOF Machado is trend-level ($p = 0.058$). Sub-09 is captured by 1-DOF Machado ($\Delta\lambda = 13.5$ nm, $\rho = 0.762$, $p = 0.018$) without cortical augmentation; the shift magnitude falls within the established range for moderate-severe protanomaly (Neitz & Neitz, 2011). The large positive cortical gain for Sub-08 ($g = +2.25$, indicating >100% overshoot beyond the exact Tregillus compensation level of $g = -1$; Tregillus et al., 2021) is not physiologically interpretable as a gain parameter and is treated as an empirical characterization of residual hV4 structure. The cross-subject model-class divergence (R+C for Sub-08; 1-DOF Machado for Sub-09) prevents a common-axis comparison of the two cases' distortion mechanisms.

Correction-vector divergence. Despite comparable LOCO fit quality ($\rho = 0.857$ for R+C vs. 0.881 for 2-component), the R+C and 2-component correction vectors for Sub-08 have cosine similarity -0.18 and sign agreement only 3/8, confirming that detection converges while correction directions diverge at the cortical level.

Δ RDM cross-model convergence. Using R+C parameters for Sub-09 ($\Delta\lambda = 19.5$ nm, $g = -1.10$), an independent Δ RDM test reaches $p = 0.026$ (40,320

cosine-similarity permutations on crossnobis distances), providing cross-model corroboration for the Sub-09 distortion signal. Retinal-family optimisation directly against Δ RDM fails for both CVD subjects; LOCO is the more discriminating phenotype and Δ RDM serves as a convergence cross-check only.

Arc-compression and LOCO vulnerability. At Sub-09’s severity, Machado arc-compression ($\Delta\lambda = 13.5$ nm) collapses three hues (c4 green, c5 cyan, c6 blue) to a single perceived angle in opponent hue space. These three collapsed hues fall within Sub-09’s LOCO vulnerability cluster (c5/c6 cyan/blue), confirming that arc-compression eliminates correction capacity precisely where Sub-09 requires it most.

Table 1

All distortion model fits at hV4 LOCO for both CVD participants. Retinal family: closed-testing (Machado first; R+C only when Machado $p > 0.05$). p -values:

$8! = 40,320$ exact permutations of colour labels on Spearman ρ .

Subject	Model	DOF	Param. 1	Param. 2	ρ	p_{perm}
Sub-08 (deutan)	2-component	2	$\beta_s = 38^\circ$	$\beta_c = -14^\circ$	0.881	0.004
Sub-09 (protan)	2-component	2	$\beta_s = 6^\circ$	$\beta_c = -22^\circ$	0.690	0.035
Sub-08 (deutan)	R+C	2	$\Delta\lambda = 2.0$ nm	$g = +2.25$	0.857	0.005
Sub-09 (protan)	Machado	1	$\Delta\lambda = 13.5$ nm	—	0.762	0.018

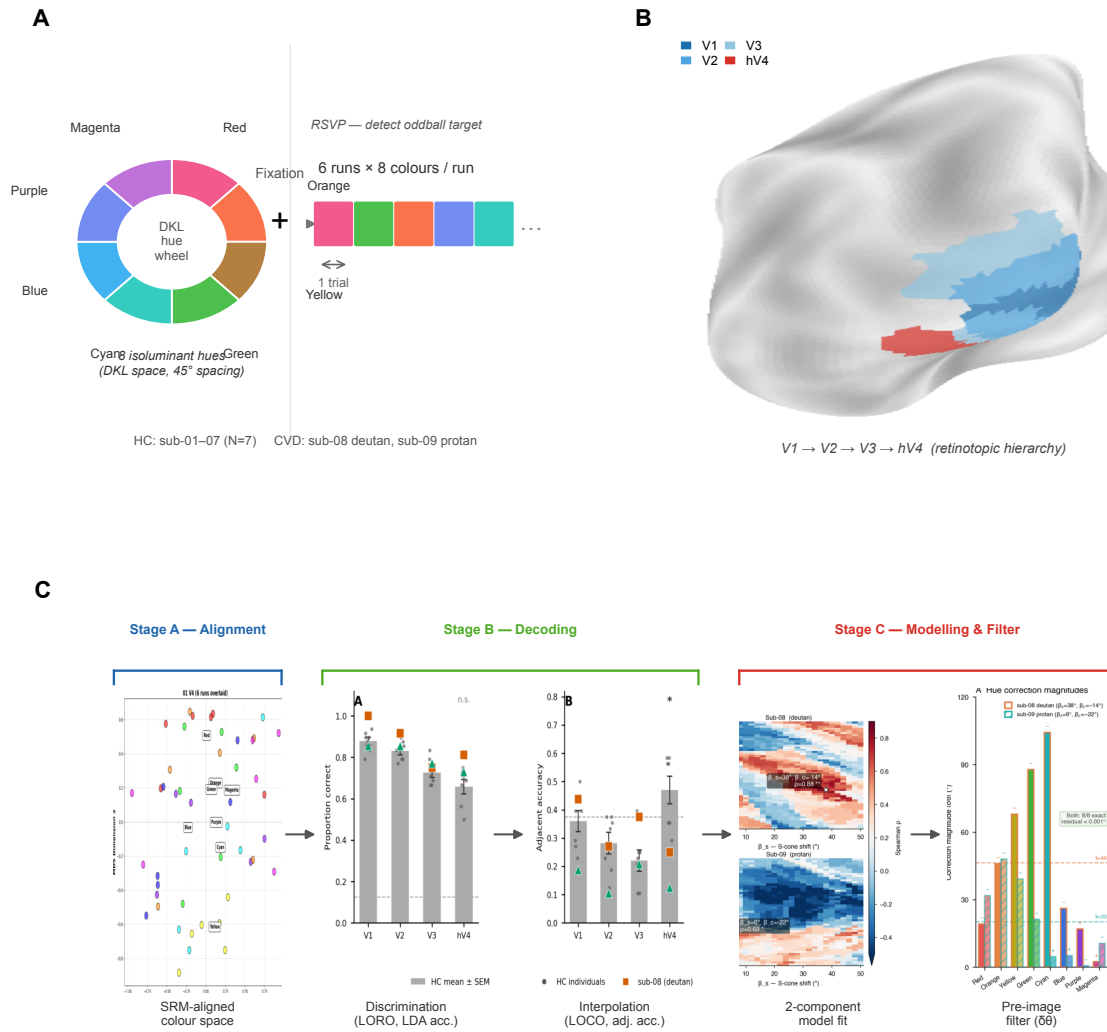


Figure 1. Experimental paradigm and analysis pipeline. (A) Stimulus set.

Isoluminant chromatic stimuli were drawn from a DKL (Derrington–Krauskopf–Lennie) hue wheel at 8 equally-spaced angles (45° apart), spanning red, orange, yellow, green, cyan, blue, purple, and magenta. **(B)** Retinotopic regions of interest (ROIs). Four areas (V1, V2, V3, hV4) were delineated on each participant’s cortical surface using standard retinotopic mapping procedures. **(C)** Analysis pipeline. Stage A (navy): raw fMRI responses were modelled with GLMs to obtain single-trial β amplitudes, aligned across sessions with Procrustes rotation, and projected into a group-shared color representation space using Shared Response Modelling (SRM; trained on HC participants only). Decoding stages (green): LORO and LOCO assessed discrimination and interpolation capacity, respectively. CVD characterisation (coral): a 2-component model decomposed individual CVD distortions into a retinal component (β_s) and a cortical component (β_c), and the pre-image of the estimated distortion yielded a stimulus-space filter specific to each CVD subject. HC: healthy controls, $n = 7$ (sub-01–07); deutan CVD: sub-08; protan CVD: sub-09.

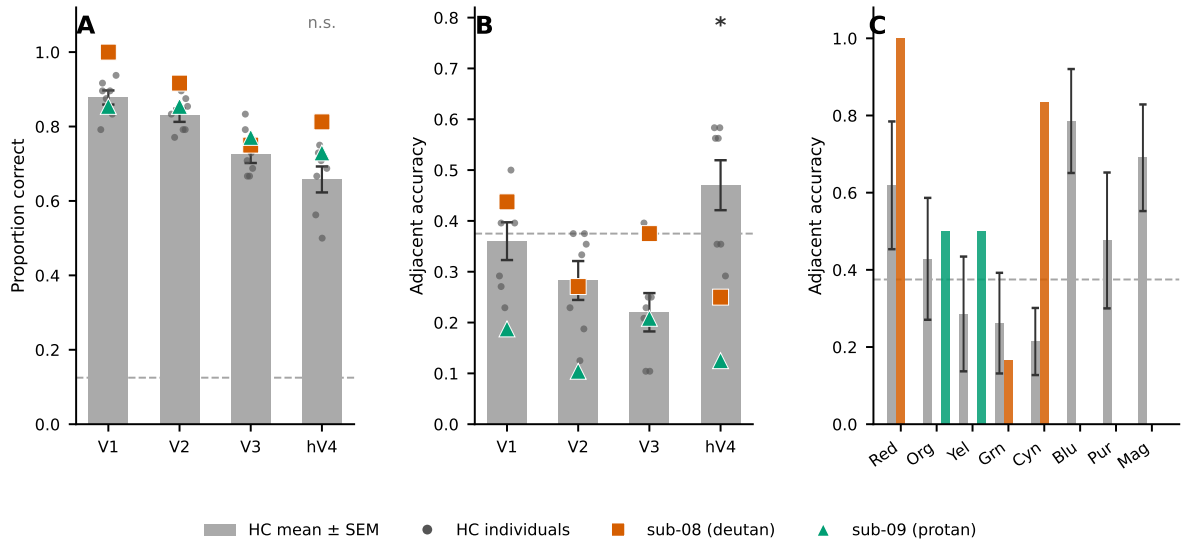


Figure 2. Color discrimination is preserved but interpolation is selectively impaired at hV4 in CVD. (A) Leave-one-run-out (LORO) discrimination accuracy (LDA, 8-class, SRM-aligned) across visual areas V1–hV4 (defined in Figure 1). Gray bars: healthy control (HC) group mean $\pm$ SEM ($n = 7$); individual HC values shown as dots. Orange square: sub-08 (deutan CVD); teal triangle: sub-09 (protan CVD). Dashed line: exact-accuracy chance ($1/8 = 0.125$). n.s.: HC-to-HC vs. HC-to-CVD cross-subject LDA generalization, all ROIs pooled (28 HC-to-HC pairs vs. 12 HC-to-CVD pairs, Mann–Whitney U , $p = 0.668$). (B) Leave-one-colour-out (LOCO) adjacent accuracy (ForwardEncoding, Procrustes-aligned). Adjacent accuracy: proportion of predictions within ± 1 hue step of the target colour (0–1; higher = better). Dashed line: adjacent-accuracy chance level ($3/8 = 0.375$). *: sub-09 falls significantly below the HC distribution at hV4 (Crawford & Howell, 1998, modified t -test, $t = -2.48$, $p = 0.024$); sub-08: $t = -1.58$, $p = 0.082$, n.s. Asterisk refers to sub-09 only. (C) Per-hue adjacent accuracy at hV4. HC bars: group mean $\pm$ SEM. Hue abbreviations: Org = orange, Yel = yellow, Grn = green, Cyn = cyan, Blu = blue, Pur = purple, Mag = magenta. Dashed line: $3/8$ chance. Per-hue Crawford & Howell tests (one-tailed, uncorrected): blue — sub-08 $d = 2.13$, $p = 0.047$; sub-09 $d = 2.15$, $p = 0.046$; purple — sub-08 $d = 2.40$, $p = 0.033$.

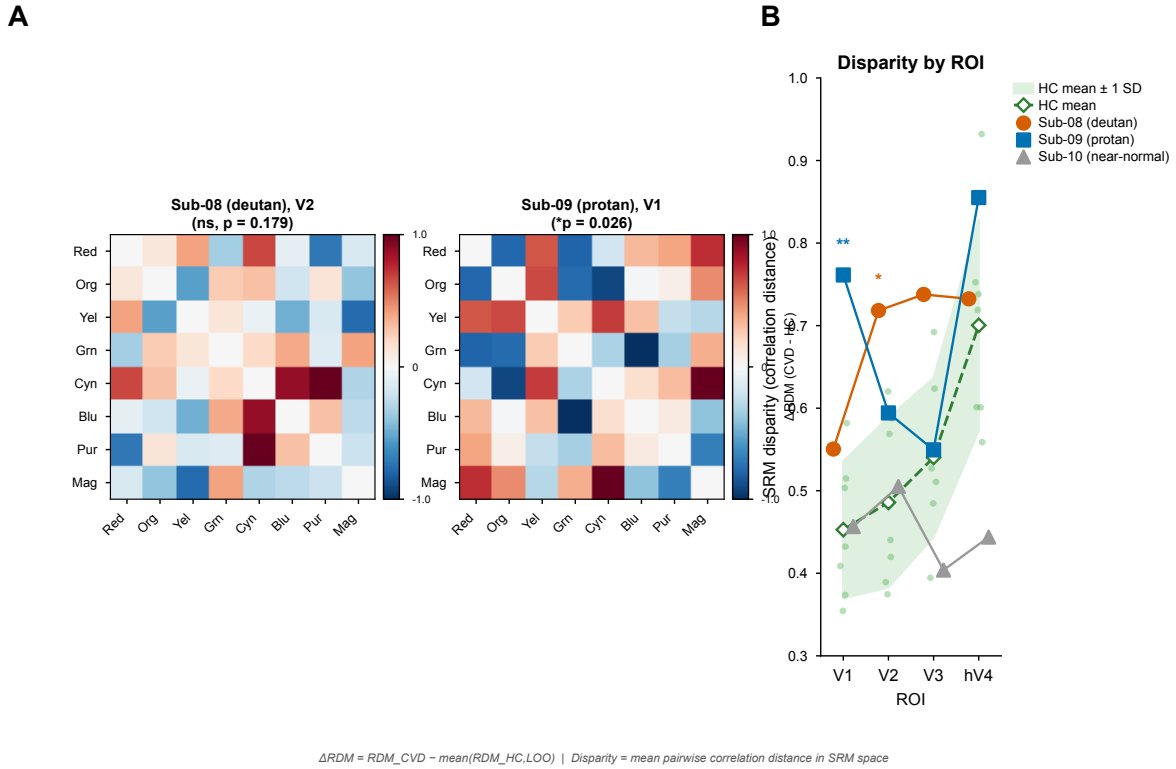


Figure 3. Each CVD subject shows significantly elevated color representation disparity at a distinct ROI. (A) Representational dissimilarity matrix (RDM) difference ($\Delta RDM = RDM_{CVD} - \overline{RDM}_{HC-LOO}$) in SRM-aligned space, for sub-08 at V2 (left) and sub-09 at V1 (right). Each subject is shown at their primary ROI, defined as the ROI with a significant pairwise-disparity elevation in (B) (V2 for sub-08, V1 for sub-09). Warm colours: CVD pairwise distances larger than HC mean; cool colours: smaller. Annotated p -values are from a permutation test of whether the retinal + cortical (R+C) cone-shift model predicts the observed ΔRDM structure (sub-08 V2: $p = 0.179$, n.s.; sub-09 V1: $p = 0.026*$). These p -values test model-geometry concordance and are independent of the disparity tests in (B). **(B)** Mean pairwise correlation distance (disparity) in SRM-aligned space per subject per ROI. HC band: group mean ± 1 SD ($n = 7$); individual HC leave-one-out values shown as dots. Significance markers from Crawford & Howell (1998) modified t -tests comparing each CVD subject to the HC distribution (sub-08 V2: $p = 0.040*$; sub-09 V1: $p = 0.007**$). Sub-10 (near-normal deutan): gray triangles, no significant elevation at any ROI. Note: (A) tests whether a cone-shift model predicts ΔRDM geometry; (B) tests whether absolute pairwise disparity is elevated — these are independent measures and their significance patterns need not agree.

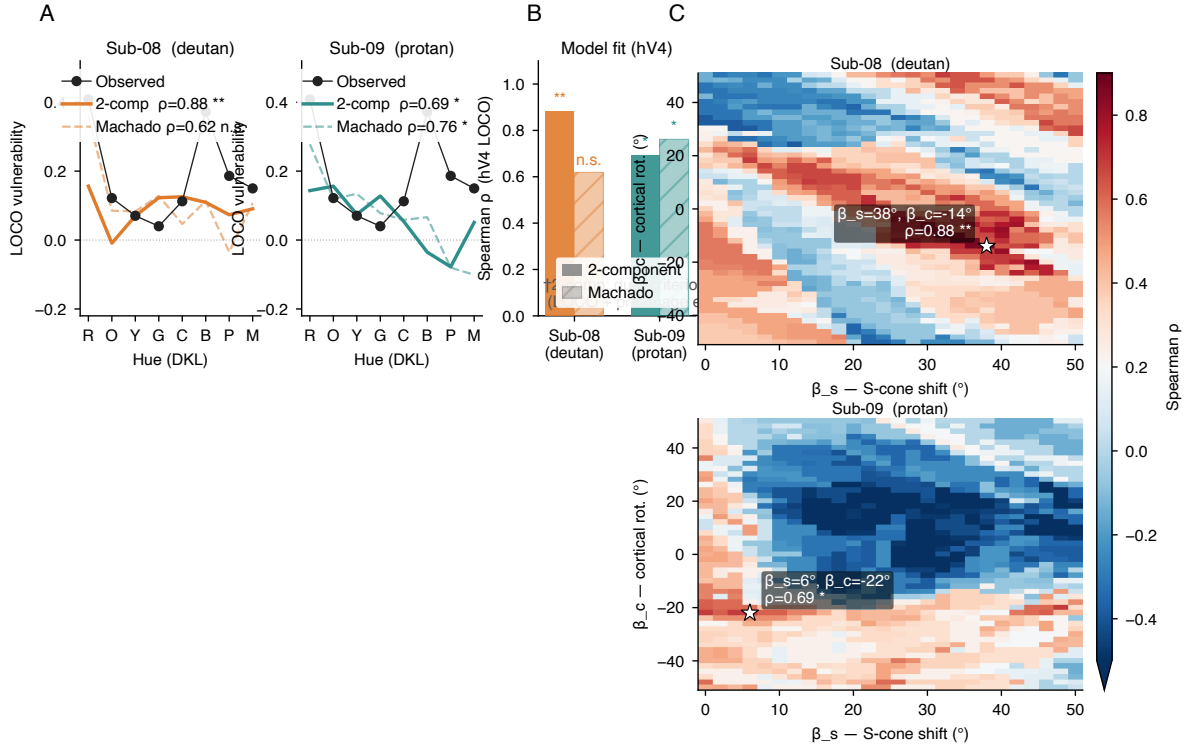


Figure 4. A two-component model (retinal shift + cortical rotation) predicts hV4 LOCO vulnerability in individual CVD subjects (sub-08: $\rho = 0.88$, $p = 0.004$; sub-09: $\rho = 0.69$, $p = 0.035$). (A) Per-hue LOCO vulnerability profiles at hV4. Filled circles: observed CVD vulnerability (HC-trained ForwardEncoding). Solid lines: 2-component model prediction at optimal (β_s, β_c) . Dashed lines: Machado (1-parameter cone-shift) prediction. X-axis hue labels (0–315°, 45° spacing in DKL hue space): R = red, O = orange, Y = yellow, G = green, C = cyan, B = blue, P = purple, M = magenta. (B) Spearman ρ between observed and model-predicted vulnerability across 8 hues. Solid bars: 2-component model; hatched bars: Machado. p -values from label-permutation test (40,320 permutations). Sub-08: 2-component $\rho = 0.88$, $p = 0.004$; Machado $\rho = 0.62$, $p = 0.058$ (n.s.). Sub-09: 2-component $\rho = 0.69$, $p = 0.035$; Machado $\rho = 0.76$, $p = 0.018$. (C) Parameter landscape: Spearman ρ as a function of β_s (S-cone retinal shift) and β_c (cortical opponent-channel rotation). White star: LOCO-optimal parameters (sub-08: $\beta_s = 38^\circ$, $\beta_c = -14^\circ$; sub-09: $\beta_s = 6^\circ$, $\beta_c = -22^\circ$). Colormap: RdBu_r (blue = low ρ ; red = high ρ).

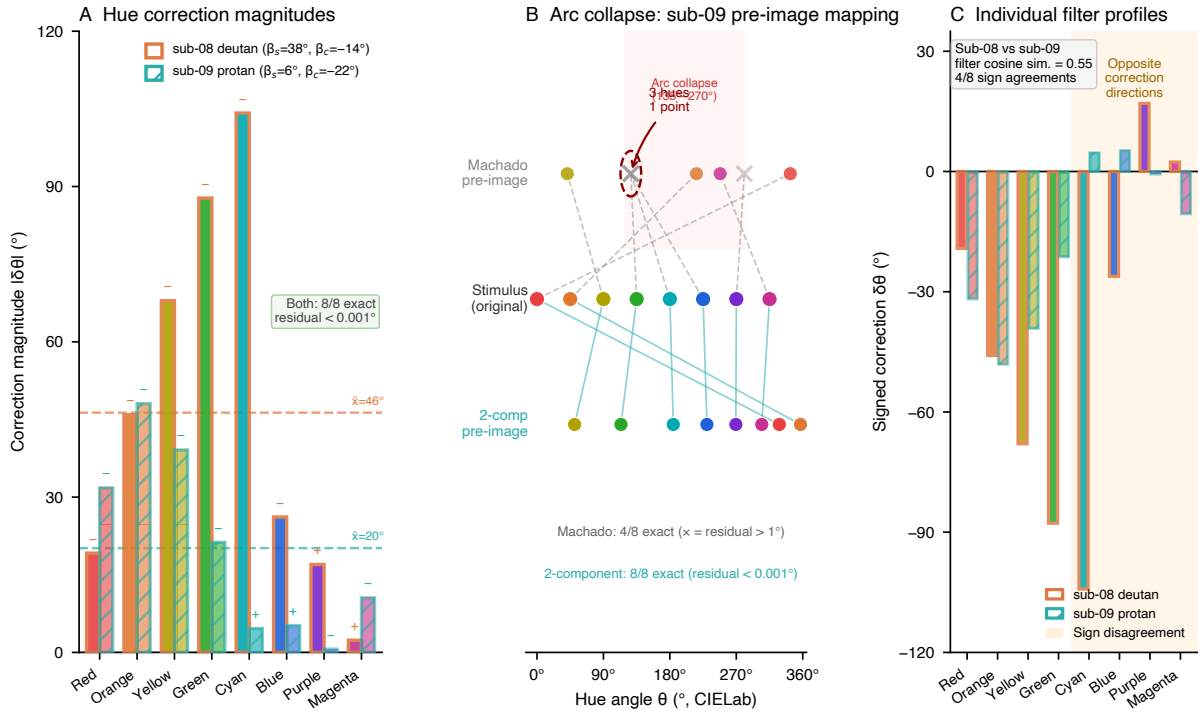


Figure 5. The 2-component pre-image filter is exact for both CVD subjects and individually distinct. (A) Hue correction magnitude ($|\delta\theta|$) at each stimulus position, computed as the exact pre-image of the 2-component distortion. Bar face colour: approximate perceptual hue of each stimulus. Dashed lines: per-subject mean correction (sub-08: 46.3° ; sub-09: 20.1°). Both subjects: 8/8 pre-images exact (maximum residual $< 0.001^\circ$). **(B)** Arc collapse in the Machado pre-image for sub-09. Green (135°), cyan (180°), and blue (225°) stimuli all map to a single pre-image angle ($\sim 127^\circ$), preventing exact inversion (4/8 exact; $\times$ marks failed hues). The 2-component pre-image maps each stimulus to a distinct corrected position across the full hue circle (8/8 exact). **(C)** Signed correction profiles ($\delta\theta$) for sub-08 (orange) and sub-09 (teal). Shaded region: sub-08 and sub-09 corrections have opposite signs (hues 5–8, cyan–magenta arc; 4/8 sign agreements overall; sub-08 vs. sub-09 filter cosine similarity = 0.55).